

AI at Work: Why GenAI is More Likely to Support Workers Than Replace Them

Of the more than 2,800 unique work skills identified and assessed by Indeed, none are “very likely” to be replaced by generative artificial intelligence.

By Annina Hering, Ph.D. & Arcenis Rojas



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Indeed's AI at Work is a series of research reports from [Hiring Lab](#), Indeed's economic research arm. This research is designed to provide data and new insights into the impact of AI on the labor market to help employers and job seekers better understand and prepare for the changing workforce.

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Annina is a senior economist at Indeed Hiring Lab and was the principal researcher for *Why GenAI is More Likely to Support Workers Than Replace Them*. Annina wrote her dissertation in the social science at the University of Cologne. Previously she worked as a postdoctoral researcher at the Max Planck Institute for the Study of Societies, where she was also part of the International Max Planck Research School on the Social and Political Constitution of the Economy (IMPRS-SPCE). She holds a M.Sc. in Sociology and Empirical Research from the University of Cologne and a B.A. in Politics and Society from the University of Bonn.



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Arcenis is a data scientist at Hiring Lab. He has a Bachelor's in Applied Economics from Ithaca College and a Master's in Economics from the City University of New York — the city college where he found a love for working with economic data. Arcenis previously worked at the U.S. Bureau of Labor Statistics where he gained proficiency in programming in R and coordinated activities for the R Users' Group. Later he went on to work as a data science consultant for U.S. federal agencies.



The Indeed Hiring Lab assessed the ability of GPT-4o, a GenAI model developed by OpenAI, to perform more than 2,800 job skills. Each skill was assessed across three main areas: The ability of GenAI to provide theoretical knowledge related to the skill; The ability of GenAI to solve problems using the skill; and GenAI's determination of the importance of physical presence in utilizing that skill. These three ratings were considered by the model as part of an ultimate assessment of the likelihood that GenAI could replace a human in performing any of those 2,800+ individual skills.

KEY POINTS

- Out of more than 2,800 work skills assessed by Indeed, none were deemed “very likely” to be replaced by GenAI. Out of five possible outcomes (very unlikely, unlikely, possible, likely, very likely), the majority (68.7%) of skills assessed were “very unlikely” or “unlikely” to be replaced by GenAI.
 - Roughly a quarter of skills (28.5%) could possibly be replaced by GenAI in the future if businesses change some practices and the tools improve.
- Overall, GenAI is generally strong at providing theoretical knowledge of skills but less so at using skills to solve problems. And as long as a skill requires significant hands-on execution (for example, “aviation” or “cooking” skills), the usefulness of GenAI will remain limited.
- Ultimately, the impact of GenAI on a given occupation will depend on the skills in that occupation that can or cannot be performed well by the technology, both today and in the future.
 - In jobs with a high share of skills that require hands-on execution, including nursing, GenAI could help with some repetitive tasks (like documentation) and allow workers to refocus on the core skills necessary in these roles.
 - In more stereotypical “office jobs,” including software development, GenAI may potentially be able to offer significant knowledge and solve modest problems, emphasizing the importance of continued upskilling and ongoing learning for human workers.